PART 1: PROJECT OVERVIEW, EXECUTIVE SUMMARY & BUSINESS OBJECTIVES

Project Title: Jumia-Style Solar E-Commerce Platform
Company: Skysoft Systems
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Development Period: 15 Days

1.1 Introduction

The proposed project involves the design, development and deployment of a professional solar e-commerce platform inspired by the shopping experience of Jumia. The platform will enable customers to browse, compare, order and purchase solar products online while allowing the business to manage its products, inventory, sales, customers, payments and business documents from a centralised administration system.

The platform will be developed by Skysoft Systems to serve as both an online shopping website and an internal business management solution. Rather than operating separate systems for online sales, physical sales, inventory and financial documentation, the business will manage these operations through one integrated platform.

The system will support a wide range of solar equipment, including:

- **Solar panels**
- **Solar inverters**
- **Solar batteries**
- **Solar water pumps**
- **Solar streetlights and security lights**
- **Charge controllers**
- **Solar cables and connectors**
- **Mounting structures and installation accessories**
- **Complete solar power systems**

The platform will provide a modern, mobile-friendly interface that allows customers to find products, examine specifications, check availability, add items to their shopping carts and place orders.

On the business side, authorised staff will be able to manage inventory, process orders, generate quotations, issue invoices and receipts, monitor sales and review business performance.

The ultimate goal is to establish a reliable digital sales channel that supports business growth, improves operational efficiency and enhances the customer experience.

1.2 Executive Summary

Skysoft Systems proposes to develop a comprehensive solar e-commerce platform that combines online shopping functionality with inventory management, sales processing, customer management and automated business-document generation.

The platform will feature a professional storefront with product categories, search functionality, product filters, comparison tools, product images, technical specifications, pricing, stock availability and customer accounts.

Customers will be able to add products to their carts, submit orders, select available payment methods and track order progress. The system will also provide access to relevant order documents, including invoices and receipts.

The administrative side will include a dashboard for managing products, inventory, suppliers, customers, orders, quotations, invoices, receipts and sales reports. Different user roles will have defined permissions to protect sensitive information and ensure accountability.

An integrated inventory module will track stock quantities, record stock movements, identify low-stock products, support multiple business locations and update stock according to configured sales and order-processing rules.

The platform will generate professional business documents containing the company's branding and relevant business information, including its KRA PIN and applicable VAT details.

The proposed technology stack consists of a React and Tailwind CSS frontend, PHP Laravel backend API, PostgreSQL database and DomPDF

document-generation functionality. The final hosting environment must support the selected backend and database technologies.

The project is scheduled for completion within 15 days, subject to timely receipt of the required client materials, approvals and access credentials. The completed platform will provide the business with a unified digital environment for managing online sales and essential business operations.

1.3 Background and Business Problem

Many businesses selling solar equipment face operational challenges when their sales, inventory and documentation processes are managed separately. These challenges can affect customer satisfaction, staff productivity, stock accuracy and management's ability to make informed decisions.

A. Limited Online Visibility

Customers increasingly search online before purchasing equipment. A business without a professional e-commerce website may struggle to reach customers beyond its existing physical market.

Proposed solution: Develop an online storefront where customers can discover products, review technical specifications, compare alternatives and submit orders at any time.

B. Manual Inventory Management

When stock is tracked using notebooks or disconnected spreadsheets, quantities can become inaccurate. Staff may also struggle to determine which products are available at different business locations.

Proposed solution: Implement a centralised inventory system that records stock received, stock sold, stock transferred, damaged products and other stock movements.

C. Manual Quotation and Invoice Preparation

Preparing quotations, invoices and receipts manually consumes time and increases the likelihood of calculation errors, duplicate references and inconsistent document formatting.

Proposed solution: Introduce automated document generation with unique numbering, automatic calculations and consistent company branding.

D. Poor Visibility into Sales Performance

Without consolidated reporting, management may find it difficult to determine total sales, pending orders, outstanding payments and the products generating the most revenue.

Proposed solution: Provide a dashboard and reports that consolidate information from sales, orders, payments and inventory.

E. Fragmented Customer Records

When customer details and purchase history are stored in different places, staff may struggle to respond to enquiries or retrieve previous transactions.

Proposed solution: Maintain customer profiles linked to orders, invoices, receipts and relevant support interactions.

F. Difficulty Managing Multiple Locations

A business operating stores or warehouses in locations such as Nairobi and Machakos needs to know where its products are held and how stock moves between locations.

Proposed solution: Introduce multi-location inventory records and controlled stock-transfer workflows.

1.4 Project Aim

The primary aim is to develop a professional, secure and scalable solar e-commerce platform that enables the business to sell products online while integrating inventory management, order processing, customer management, payment recording and business-document generation into one system.

The solution will support both online transactions and authorised physical or office sales, allowing the business to maintain consistent records across its sales channels.

1.5 Specific Project Objectives

The project will pursue the following objectives.

Objective 1: Establish an Online Sales Channel

Develop a modern e-commerce website through which customers can browse solar products, search for specific items, compare alternatives and place orders.

Objective 2: Improve Product Presentation

Provide structured product pages containing images, descriptions, technical specifications, prices, availability and downloadable datasheets where available.

Objective 3: Centralise Inventory Management

Maintain accurate stock records, record stock movements, provide low-stock alerts and support transfers between configured locations.

Objective 4: Automate Business Documents

Generate quotations, invoices and receipts using standard templates, automatic calculations, unique reference numbers and the company's branding.

Objective 5: Streamline Order Processing

Provide a structured process for receiving, confirming, preparing, dispatching and completing orders.

Objective 6: Improve Customer Management

Maintain customer accounts and transaction histories, allowing customers to access their orders and relevant documents while helping staff respond efficiently to enquiries.

Objective 7: Support Payment Processing

Provide the agreed payment options, including M-Pesa Till instructions, bank payment and Cash on Delivery where offered. Automated M-Pesa STK Push may be incorporated as an additional integration, subject to the agreed scope and access to the required payment services.

Objective 8: Strengthen Management Reporting

Provide dashboards and reports covering sales, orders, inventory, product performance and other operational indicators.

Objective 9: Improve Access Control and Accountability

Establish separate permissions for the Super Admin, Inventory Manager, Sales Agent and Customer, with important administrative activities recorded in an audit trail.

Objective 10: Improve Online Discoverability

Implement search-engine optimisation practices, including appropriate page metadata, product URLs, structured product information and server-side rendering or pre-rendering where required by the final architecture.

1.6 Project Scope

The project will cover the following principal functional areas.

A. E-Commerce Storefront

- **Product catalogue and categories**
- **Product search and filters**
- **Product comparison**
- **Shopping cart and checkout**
- **Customer registration and accounts**
- **Order tracking**
- **Responsive mobile and desktop interfaces**

B. Inventory Management

- **Product and stock records**
- **Stock receiving and deductions**
- **Low-stock alerts**
- **Supplier management**
- **Stock adjustments and movement history**
- **Multi-location inventory and stock transfers**

C. Sales and Business Documents

- **Order management**
- **Quotation generation**

- **Invoice and receipt generation**
- **Automatic document numbering**
- **PDF downloads**
- **Sales dashboards and reports**

D. Customer Management

- **Customer profiles**
- **Purchase history**
- **Saved addresses**
- **Wishlists and product reviews**
- **Access to personal orders and documents**

E. Administration and Security

- **Role-based access control**
- **Company settings**
- **Activity logs**
- **Operational dashboards**
- **Data-protection measures**

Features will be implemented according to the agreed requirements and the capabilities of the selected hosting environment. Any additional features beyond the approved scope will be assessed and agreed upon before implementation.

1.7 Target Users

The system will serve four primary user groups.

User	Main Responsibility
Super Admin / Business Owner	**Controls system settings, users, financial visibility, business records and overall operations.**

Inventory Manager	**Manages products, suppliers, stock, stock movements and relevant order-fulfilment activities.**
Sales Agent	**Handles customer enquiries, quotations, invoices, sales and assigned order-processing tasks.**
Customer	**Browses products, places orders, tracks purchases and accesses personal transaction documents.**

Each user will have permissions appropriate to their responsibilities. Customers will not have access to internal inventory costs, supplier records or other customers' private information.

1.8 Proposed Technology and Architecture

The platform will use a modern web architecture that separates the customer interface, business logic and data storage.

Component	Proposed Technology	Purpose
Frontend	**React**	**Customer storefront and administrative interfaces**
Styling	**Tailwind CSS**	**Responsive layouts and consistent design**
Backend	**PHP Laravel**	**Business rules, authentication, permissions and API services**

Database	**PostgreSQL**	**Products, users, orders, inventory and transaction records**
PDF Generation	**DomPDF**	**Quotations, invoices and receipts**
Security	**HTTPS and server-side access controls**	**Protects data and restricts unauthorised actions**
Hosting	**Compatible PHP/PostgreSQL hosting**	**Makes the platform available online**

The React frontend will communicate with the Laravel backend through an API. The backend will validate requests, apply business rules, enforce user permissions and interact with the PostgreSQL database.

This architecture will help ensure that stock levels, orders and customer records remain consistent across the storefront and administrative system.

1.9 Expected Project Deliverables

The project is expected to deliver:

1. **A responsive solar e-commerce website.**
2. **A product catalogue and category management module.**
3. **Customer registration, login and account management.**
4. **Shopping cart, checkout and order management.**
5. **An inventory management module.**
6. **Supplier and stock-movement management.**
7. **A role-based administrative dashboard.**
8. **Quotation, invoice and receipt generation.**
9. **PDF document downloads.**
10. **Agreed payment options and payment-recording functionality.**
11. **Sales and inventory reports.**

12. **Search-engine optimisation foundations.**
13. **Security and access-control implementation.**
14. **Testing and deployment to the agreed hosting environment.**
15. **Basic operational handover and documentation.**
16. **Three months of post-launch support under the proposed terms.**

The final deliverables will be verified against the agreed project scope during testing and handover.

1.10 Project Timeline — 15 Days

The project will be implemented over a period of 15 days, commencing after the initial project arrangements have been completed and the required client materials have been provided.

Period	Activities
Days 1–2	Requirements confirmation, project setup, architecture and database design
Days 3–6	Storefront, product catalogue, search, customer accounts and core e-commerce functionality
Days 7–10	Inventory, order processing, business documents and administrative dashboard
Days 11–12	Integrations, reporting, SEO configuration and responsive improvements

Days 13–14	**Testing, corrections, security checks and user acceptance**
Day 15	**Deployment, final verification and handover**

Timely provision of product data, images, branding, company details, approvals and hosting access will be essential to achieving this schedule. Delays in client feedback, third-party services or hosting configuration may affect delivery.

1.11 Client Responsibilities

To support timely development, the client will provide the necessary business information and materials, including:

- **Company logo and branding**
- **Business name and contact information**
- **KRA PIN and applicable VAT details**
- **Product names and descriptions**
- **Selling prices and available product quantities**
- **Product images and technical datasheets**
- **Supplier information, where required**
- **M-Pesa Till details**
- **Delivery and payment policies**
- **Terms and conditions**
- **Hosting and domain access, where applicable**
- **Feedback and approvals during development**

The client is responsible for ensuring that the supplied business, tax and product information is accurate and authorised for use.

1.12 Security, Maintenance and Support

The platform will incorporate appropriate security measures, including secure authentication, server-side role enforcement, HTTPS configuration and controls for accessing customer and business records.

Where supported by the hosting environment, regular backups will be configured to help protect business data. Backup arrangements and restoration procedures will be confirmed during deployment.

The proposed agreement includes three months of post-launch support for issues covered by the agreed scope. New features, substantial modifications, third-party service charges and ongoing hosting or domain renewals will be handled according to the agreed support and maintenance arrangements.

1.13 Expected Business Benefits

The proposed system is expected to provide the following benefits:

- **Wider market reach: Customers can discover products online beyond the business's immediate physical location.**
- **Improved stock accuracy: Centralised inventory records reduce reliance on disconnected spreadsheets.**
- **Faster sales processing: Orders and documents can be handled through one platform.**
- **Professional documentation: Consistent quotations, invoices and receipts improve business presentation.**
- **Better customer service: Customers can access product information and track orders.**
- **Improved management visibility: Reports help management review sales and operational performance.**
- **Stronger accountability: User permissions and activity logs support traceability.**
- **Support for business growth: The platform provides a foundation for additional products, locations and integrations.**

These benefits depend on effective implementation, accurate data, staff adoption and ongoing maintenance.

1.14 Conclusion

The Jumia-Style Solar E-Commerce Platform will provide a unified digital solution for online sales and the supporting business operations required to run a solar equipment enterprise.

By integrating the storefront, inventory, customer records, order processing, payments and business documents, the platform will help the business move

from fragmented manual processes toward a more structured and efficient operating model.

Skysoft Systems proposes to complete the agreed platform within 15 days, subject to timely receipt of the required information, approvals and fulfilment of the project assumptions.

The resulting platform will provide a professional digital sales channel and an integrated business management environment designed to improve customer service, strengthen operational control and support future growth.

Add a clear payment-term placeholderClarify the 15-day delivery assumptions

2. System Features

The proposed solar e-commerce platform will combine a modern online shopping experience with a complete business management system. The platform will serve two major purposes: selling solar products online and managing the day-to-day operations of the solar business.

The system will be divided into the following major modules:

2.1 Customer-Facing E-Commerce Storefront

The storefront is the public-facing section of the platform where customers interact with the business.

It will provide a modern shopping experience inspired by leading e-commerce platforms such as Jumia, while being specifically designed for solar products.

Homepage

The homepage will act as the main entry point to the online store.

It will include:

- **Company branding and logo.**
- **Main navigation menu.**
- **Search bar.**

- **Promotional banners.**
- **Featured solar products.**
- **Flash deals and discounted products.**
- **New arrivals.**
- **Best-selling products.**
- **Recommended products.**
- **Product categories.**
- **Special offers.**
- **Customer reviews.**
- **Contact information.**
- **WhatsApp customer-support button.**
- **Links to social media pages.**
- **Footer containing company information and important links.**

The homepage will be designed to immediately communicate what the business sells and direct customers towards products and offers.

2.2 Product Categories

Products will be organised into structured categories to make navigation easier.

Initial categories will include:

Solar Panels

Products can be organised according to:

- **Wattage.**
- **Brand.**
- **Panel type.**
- **Price.**
- **Availability.**

Examples:

- **100W Solar Panels**
- **200W Solar Panels**
- **300W Solar Panels**
- **450W Solar Panels**
- **550W Solar Panels**

Solar Inverters

The system can support:

- **Hybrid inverters.**
- **Off-grid inverters.**
- **Grid-tie inverters.**
- **Pure sine wave inverters.**
- **Different power ratings.**

Solar Batteries

Products can include:

- **Lithium batteries.**
- **Gel batteries.**
- **AGM batteries.**
- **Tubular batteries.**
- **Other supported battery technologies.**

Solar Water Pumps

Customers can browse pumps based on:

- **Pump capacity.**
- **Voltage.**
- **Power rating.**
- **Application.**
- **Brand.**
- **Price.**

Solar Lighting

This category may include:

- **Solar security lights.**
- **Solar floodlights.**
- **Solar street lights.**
- **Solar home lighting systems.**
- **Outdoor solar lights.**

Solar Accessories

Examples include:

- **Charge controllers.**
- **Solar cables.**
- **Connectors.**
- **Mounting structures.**
- **Fuses.**
- **Breakers.**
- **MC4 connectors.**
- **Solar installation accessories.**

The administrator will be able to create additional categories when required.

2.3 Product Management

Every product will have its own dedicated product record.

The administrator will be able to enter information such as:

- **Product name.**
- **SKU.**
- **Barcode.**
- **Category.**
- **Brand.**
- **Product description.**
- **Selling price.**
- **Discount price.**
- **Buying price.**
- **Stock quantity.**
- **Minimum stock level.**
- **Product images.**
- **Technical specifications.**
- **Datasheet PDF.**
- **Warranty information.**
- **Product status.**
- **Supplier.**
- **Store/location.**
- **Featured-product status.**

This allows the business to maintain a centralised product catalogue.

Product Availability

Each product can automatically display statuses such as:

- **In Stock**
- **Low Stock**
- **Out of Stock**
- **Coming Soon**

This prevents customers from ordering products that are unavailable.

2.4 Product Details Page

Each product will have a professional product page containing detailed information.

The page will display:

- **Product name.**
- **Product images.**
- **Product price.**
- **Original price where discounted.**
- **Discount percentage.**
- **Stock availability.**
- **Product description.**
- **Technical specifications.**
- **Brand.**
- **SKU.**
- **Warranty.**
- **Datasheet download.**
- **Quantity selector.**
- **Add to Cart button.**
- **Add to Wishlist button.**
- **Compare button.**
- **Related products.**
- **Customer ratings.**
- **Customer reviews.**

For technical solar products, the specifications section will be particularly important.

For example, a solar panel could display:

Specification	Example
Rated Power	550W
Maximum Voltage	41.8V
Maximum Current	13.16A
Cell Type	Monocrystalline
Warranty	12 Years
Brand	Example Brand

2.5 Smart Search

The system will include a powerful search engine allowing customers to quickly locate products.

Customers could search for:

- **"550W solar panel"**
- **"solar battery"**
- **"5kW inverter"**
- **"solar water pump"**
- **"solar lights"**
- **"lithium battery"**

Search results will display relevant products rather than requiring customers to manually browse every category.

The search system can also support product names, SKUs and selected technical attributes.

2.6 Advanced Product Filtering

Customers will be able to narrow down products using filters.

Depending on the category, filters may include:

- **Price range.**
- **Brand.**
- **Wattage.**
- **Product category.**
- **Availability.**
- **Product type.**
- **Battery capacity.**
- **Voltage.**
- **Inverter capacity.**
- **Pump capacity.**

For example, a customer looking for solar panels could select:

Category: Solar Panels
Power: 400W–600W
Brand: Selected Brand
Price: KSh 20,000–40,000
Availability: In Stock

The system will immediately display matching products.

2.7 Product Comparison

Customers will be able to compare up to four products simultaneously.

The comparison screen can display:

- **Product image.**
- **Product name.**
- **Price.**
- **Brand.**
- **Wattage.**
- **Voltage.**
- **Technical specifications.**
- **Warranty.**
- **Availability.**

This is especially useful for technical solar products where customers need to compare specifications before purchasing.

2.8 Shopping Cart

Customers will have a shopping cart where they can manage selected products.

The cart will show:

- **Product name.**
- **Product image.**
- **Unit price.**
- **Quantity.**
- **Subtotal.**
- **Discounts.**
- **Total amount.**

Customers can:

- **Increase quantity.**
- **Reduce quantity.**
- **Remove products.**
- **Continue shopping.**
- **Proceed to checkout.**

The system will calculate totals automatically.

2.9 Wishlist

Registered customers will be able to save products for future purchases.

A customer can add a product to their wishlist without immediately purchasing it.

The wishlist can be used for:

- **Products they intend to buy later.**
- **Products they are comparing.**
- **Products awaiting funds.**
- **Products they want to monitor.**

2.10 Customer Registration and Account

Customers will be able to create accounts on the platform.

Customer accounts will store:

- **Name.**
- **Phone number.**
- **Email address.**
- **Delivery information.**
- **Order history.**
- **Wishlist.**
- **Reviews.**
- **Saved information.**

The account dashboard will provide customers with a central place to manage their shopping activities.

2.11 Checkout System

The checkout process will be designed to be simple and mobile-friendly.

The customer will confirm:

1. **Products.**
2. **Quantities.**
3. **Delivery details.**

4. **Contact information.**
5. **Payment method.**
6. **Order information.**

The system will then generate an order reference number.

2.12 Payment Options

The platform will support multiple payment methods.

M-Pesa

The system will display the business M-Pesa Till/payment details.

Customers can make payment and provide the relevant transaction information where required.

Bank Payment

The business bank details can be displayed during checkout.

Cash on Delivery

Where the business allows it, customers can select Cash on Delivery.

Optional STK Push

Automatic M-Pesa Daraja STK Push can be integrated as an optional feature.

This will allow a customer to enter their phone number and receive an M-Pesa payment prompt directly on their phone.

Optional STK Push integration: KES 10,000.

2.13 Order Management

Every successful order will receive a unique order number.

The order management system will allow staff to monitor orders through different stages.

Example:

Pending → Confirmed → Processing → Packed → Dispatched → Delivered

Other statuses can include:

- **Cancelled.**
- **Rejected.**
- **Returned.**
- **Refunded.**

Staff will be able to see:

- **Customer details.**
- **Products ordered.**
- **Quantities.**
- **Total amount.**
- **Payment status.**
- **Delivery details.**
- **Order date.**
- **Order status.**

2.14 Order Tracking

Customers will be able to monitor their orders from their accounts.

For example:

Order #SKY-2026-00125

✓ Order Received
✓ Payment Confirmed
✓ Processing
✓ Packed
→ Out for Delivery
○ Delivered

This gives customers transparency without requiring them to call the business for every update.

2.15 Inventory Management

Inventory management will be integrated directly into the e-commerce system.

When a product is sold, its available stock will be updated.

For example:

550W Solar Panel

Opening stock: 20
Units sold: 3
Remaining stock: 17

This eliminates the need to maintain a separate spreadsheet for basic stock tracking.

2.16 Multi-Location Inventory

The system will support multiple business locations.

For example:

- **Nairobi Store**
- **Machakos Store**
- **Kitui Store**
- **Main Warehouse**

The administrator can see how much stock exists at each location.

Example:

Product	Nairobi	Machakos	Kitui	Total
550W Panel	20	15	8	43

5kW Inverter	**5**	**3**	**2**	**10**
10kWh Battery	**3**	**4**	**1**	**8**

2.17 Stock Transfers

Staff will be able to transfer stock between authorised locations.

Example:

Machakos Store → Nairobi Store

Product: 550W Solar Panel
Quantity: 5

The system will automatically:

- **Reduce Machakos stock by 5.**
- **Increase Nairobi stock by 5.**
- **Record the transfer.**
- **Record the user responsible.**
- **Record the date and time.**

2.18 Low-Stock Alerts

The system will automatically monitor inventory levels.

The initial default threshold can be set to 5 units.

For example:

> **⚠ Low Stock Alert**
> **100W Solar Panel — Only 3 units remaining.**

The administrator can then take action before the product completely runs out.

2.19 Stock Movement History

Every stock movement will be recorded.

The system can record:

- **Stock received.**
- **Stock sold.**
- **Stock transferred.**
- **Stock adjusted.**
- **Damaged stock.**
- **Lost stock.**
- **Returned stock.**

Each transaction will include:

- **Product.**
- **Quantity.**
- **Location.**
- **User.**
- **Date.**
- **Reason/reference.**

This creates accountability and makes it easier to identify discrepancies.

2.20 Supplier Management

The system will contain a supplier management module.

Supplier records can include:

- **Supplier name.**
- **Contact person.**
- **Phone number.**
- **Email.**
- **Location.**

- **Products supplied.**
- **Buying prices.**
- **Supplier notes.**

This gives the business a central supplier database.

2.21 Buying Price and Selling Price Management

The system can store both:

Buying Price
The amount the business pays the supplier.

Selling Price
The amount charged to the customer.

For example:

Buying price: KSh 28,000
Selling price: KSh 35,000

The system can use this information to calculate relevant gross margin information for authorised users.

Sensitive buying-price and profit information will be restricted to authorised roles.

2.22 Damaged, Lost and Returned Stock

The system will allow authorised users to record stock that cannot be sold normally.

Examples:

- **Damaged panel.**
- **Faulty inverter.**
- **Lost accessory.**
- **Customer return.**

- **Warranty return.**

The system will record the reason and quantity so that physical stock and system stock remain aligned.

2.23 Quotations

Staff will be able to create professional Quotations directly from the system.

A quotation can include:

- **Company logo.**
- **Company name.**
- **Contact information.**
- **KRA PIN.**
- **Customer details.**
- **Quotation number.**
- **Date.**
- **Validity period.**
- **Product description.**
- **Quantity.**
- **Unit price.**
- **Discount.**
- **VAT.**
- **Total amount.**
- **Terms and conditions.**

The quotation will be generated as a professional PDF.

2.24 Invoice Generation

Approved quotations can be converted into invoices.

The invoice will automatically contain:

- **Invoice number.**
- **Customer details.**
- **Product details.**

- **Quantity.**
- **Unit price.**
- **Discount.**
- **VAT.**
- **Total amount.**
- **Payment status.**
- **Company information.**
- **KRA PIN.**
- **Logo.**

This reduces repetitive manual data entry.

2.25 Receipt Generation

Once payment has been confirmed, staff can generate a receipt.

The receipt will include:

- **Receipt number.**
- **Customer.**
- **Invoice reference.**
- **Amount paid.**
- **Payment method.**
- **Date.**
- **Transaction/reference number where applicable.**
- **Company information.**

2.26 PDF Documents

Quotations, invoices and receipts will be downloadable as PDF documents.

The system will provide a consistent professional design for all business documents.

Documents can be:

- **Downloaded.**
- **Printed.**
- **Shared through WhatsApp.**

- **Sent by email.**

2.27 Sales Dashboard

The administration dashboard will provide an overview of business performance.

Possible dashboard indicators include:

- **Total sales.**
- **Today's sales.**
- **Weekly sales.**
- **Monthly sales.**
- **Number of orders.**
- **Pending orders.**
- **Completed orders.**
- **Low-stock products.**
- **Number of customers.**
- **Top-selling products.**

Charts will provide a visual representation of business performance.

2.28 Customer Management

The business will have a central customer database.

Staff can view:

- **Customer name.**
- **Phone number.**
- **Email.**
- **Orders.**
- **Purchase history.**
- **Delivery information.**
- **Account status.**

This allows the business to understand its customers and provide better service.

2.29 Product Reviews and Ratings

Customers can review products after purchase.

The system can display:

★★★★★

Average rating: 4.7/5

Reviews can help future customers make purchasing decisions.

Administrators can moderate inappropriate reviews according to the agreed business rules.

2.30 WhatsApp Integration

A WhatsApp contact button will be available throughout the storefront.

Customers can use it to:

- **Ask about products.**
- **Request assistance.**
- **Ask about delivery.**
- **Make sales enquiries.**
- **Contact the business directly.**

The system can also support sharing generated quotations, invoices and receipts through WhatsApp.

2.31 SEO and Google Visibility

The platform will be developed with search-engine visibility in mind.

Product and category pages will be structured to target searches such as:

- **Solar Panels Kenya**
- **Solar Batteries Kenya**
- **Solar Panels Machakos**
- **Solar Equipment Nairobi**
- **Solar Water Pumps Kenya**
- **Solar Inverters Kenya**
- **Lithium Solar Batteries Kenya**

The platform will use:

- **Search-friendly URLs.**
- **Product metadata.**
- **Category metadata.**
- **Product Schema markup.**
- **Proper page titles.**
- **Meta descriptions.**
- **Structured product information.**
- **Search-engine-readable product pages.**
- **Sitemap support.**

Because a plain React SPA can present indexing challenges, the storefront will use appropriate server-side rendering or pre-rendering for important product and category pages.

2.32 Mobile-Responsive Design

The platform will be designed primarily with mobile users in mind.

Customers will be able to shop using:

- **Smartphones.**
- **Tablets.**
- **Laptops.**
- **Desktop computers.**

The interface will automatically adapt to different screen sizes.

This is particularly important because many Kenyan customers access online stores primarily through mobile phones.

2.33 Security and Access Control

Security will be implemented at both frontend and backend levels.

The system will include:

- **Secure authentication.**
- **Password protection.**
- **Role-based permissions.**
- **API-level authorisation.**
- **Protected administrative routes.**
- **Secure database access.**
- **HTTPS/SSL.**
- **Activity logging.**
- **Restricted financial information.**
- **Controlled document access.**

For example, hiding the "Buying Price" button in React will not be the only security measure. The API will also prevent unauthorised users from requesting buying-price information.

2.34 Activity and Audit Logs

Important administrative actions will be recorded.

For example:

> **Willy Mutunga**
> **Updated stock for 550W Solar Panel**
> **Quantity: +20**
> **Location: Machakos**
> **Date: 20 October 2026**
> **Time: 10:42 AM**

The system can similarly record:

- **Product changes.**
- **Price changes.**
- **Stock adjustments.**
- **Stock transfers.**
- **Order approvals.**

- **Document creation.**
- **User activity.**

This improves accountability.

2.35 Backup and Data Protection

The production system will be configured for regular database backups.

Backups help protect against:

- **Accidental deletion.**
- **Database corruption.**
- **Server problems.**
- **Human error.**
- **System failures.**

The Super Admin will have access to backup-related controls according to the hosting and deployment configuration.

2.36 Administration Settings

The Super Admin will have a dedicated settings area for managing business information.

This can include:

- **Company name.**
- **Company logo.**
- **Address.**
- **Phone number.**
- **Email.**
- **KRA PIN.**
- **VAT information.**
- **M-Pesa Till.**
- **Bank details.**
- **Invoice numbering.**
- **Receipt numbering.**
- **Quotation numbering.**

- **Low-stock threshold.**
- **Store locations.**
- **Delivery settings.**

This means the business can update important information without requiring the developer to modify the source code.

2.37 Notifications

The system can provide notifications for important events.

Examples include:

- **New order received.**
- **Payment received.**
- **Low-stock product.**
- **Order approved.**
- **Order dispatched.**
- **Order delivered.**
- **Customer registration.**
- **New quotation.**
- **Invoice generated.**

Depending on the final implementation, notifications can be displayed inside the system and/or integrated with email, WhatsApp or other approved services.

2.38 Reporting

The platform will provide business reports to authorised users.

Reports can include:

Sales Reports

- **Daily sales.**
- **Weekly sales.**
- **Monthly sales.**

- **Sales by product.**
- **Sales by category.**

Inventory Reports

- **Current stock.**
- **Low stock.**
- **Stock movement.**
- **Damaged stock.**
- **Lost stock.**
- **Returned stock.**
- **Stock by location.**

Order Reports

- **Pending orders.**
- **Completed orders.**
- **Cancelled orders.**
- **Returned orders.**

Product Reports

- **Best-selling products.**
- **Slow-moving products.**
- **Out-of-stock products.**

Sensitive profitability reports will only be available to authorised users.

2.39 Overall Business Workflow

The system will connect all these features into one workflow:

Customer searches for product
↓
Views product specifications
↓
Compares products
↓
Adds product to cart
↓
Checks out

↓
Makes/chooses payment
↓
Order is created
↓
Inventory is updated
↓
Staff processes order
↓
Quotation/Invoice/Receipt generated where applicable
↓
Product is packed and dispatched
↓
Customer tracks order
↓
Order delivered
↓
Customer reviews product

Behind this entire process, the Super Admin and Inventory Manager can monitor products, inventory, customers, orders, suppliers and business performance from the administrative dashboard.

Core Value of the System

The proposed platform therefore goes beyond being a simple online shop. It functions as a combined e-commerce, inventory, sales, order-processing and business-document management platform, allowing the solar company to manage both its online customers and physical business operations from one central system.

3. INVENTORY MANAGEMENT SYSTEM

The inventory management module will provide the business with a centralised system for controlling, monitoring and tracking all solar products from the time they are received from suppliers until they are sold, transferred, returned, damaged or otherwise removed from available stock.

The inventory system will be fully integrated with the e-commerce storefront, order management system, supplier management and sales documents. This means that the business will not need to maintain separate spreadsheets or manually update stock every time a sale occurs.

The system will maintain a single source of truth for inventory, ensuring that the quantity displayed to customers is consistent with the quantity recorded by staff.

3.1 Centralised Product Inventory

Every product listed on the platform will have an inventory record.

For example:

Product: 550W Monocrystalline Solar Panel
SKU: SOL-PNL-550-001
Buying Price: KES 28,000
Selling Price: KES 35,000
Available Stock: 25
Minimum Stock: 5
Location: Machakos Store

The inventory record will be connected to the product catalogue so that changes made by authorised staff are reflected throughout the system.

The system will maintain information such as:

- **Product name.**
- **SKU.**
- **Barcode.**
- **Category.**
- **Brand.**
- **Buying price.**
- **Selling price.**
- **Available quantity.**
- **Reserved quantity.**
- **Sold quantity.**
- **Damaged quantity.**
- **Returned quantity.**
- **Minimum stock level.**
- **Current location.**
- **Supplier.**
- **Stock status.**

3.2 Real-Time Stock Updates

The inventory system will automatically update stock when transactions occur.

For example, if the business has:

550W Solar Panels — 20 units

A customer purchases:

3 units

The system will automatically calculate:

20 − 3 = 17 units remaining

The available stock will therefore become:

> **17 units — In Stock**

Staff do not need to manually edit the stock after every online sale.

This reduces human errors and helps prevent the business from selling products that are no longer available.

3.3 Stock Reservation

Where required, the system can distinguish between available stock and reserved stock.

For example:

Total physical stock: 20

Customer A places an order for: 3

The system can show:

- **Physical stock: 20**
- **Reserved: 3**
- **Available for new orders: 17**

This is particularly useful when an order has been placed but has not yet been fully processed.

The system can release the reserved quantity if an order is cancelled according to the configured business rules.

3.4 Stock In — Receiving New Products

When the business receives products from a supplier, an authorised Inventory Manager or Super Admin can record the incoming stock.

For example:

Supplier: Solar Equipment Supplier Ltd
Product: 550W Solar Panel
Quantity Received: 30
Buying Price: KES 28,000 per unit
Location: Machakos Store
Reference: PO-2026-015

The system will increase available inventory by 30 units.

If the previous stock was:

10 units

After receiving 30:

10 + 30 = 40 units

The transaction will be recorded in the stock history.

3.5 Stock-In Documentation

Each stock receipt can have a reference number.

The system can record:

- **Supplier.**
- **Delivery date.**
- **Product.**
- **Quantity.**
- **Buying price.**
- **Total purchase value.**
- **Location.**
- **Supplier invoice/reference number.**
- **Staff member who received the stock.**
- **Notes.**

This gives the business a clear history of where its inventory came from.

3.6 Stock-Out Management

Stock can leave the inventory for several reasons, including:

- **Customer sale.**
- **Internal transfer.**
- **Damaged product.**
- **Lost product.**
- **Warranty replacement.**
- **Promotional allocation.**
- **Manual adjustment.**

The system will require the appropriate reason when stock is removed manually.

For example:

> **Product: 5kW Hybrid Inverter**
> **Quantity: 1**
> **Reason: Customer Sale**
> **Order: #SKY-2026-00451**

or:

> **Product: 550W Solar Panel**
> **Quantity: 2**
> **Reason: Damaged Stock**

This provides accountability.

3.7 Automatic Stock Deduction From Sales

The e-commerce system will be directly connected to inventory.

When an order reaches the appropriate confirmed stage, the system will deduct the purchased quantity according to the configured business workflow.

For example:

Customer orders:

- **2 × 550W Solar Panels**
- **1 × 5kW Inverter**
- **2 × Solar Cables**

The inventory system automatically updates all three products.

This prevents staff from having to manually record each sale separately.

3.8 Low-Stock Management

The system will continuously monitor stock quantities.

Each product can have a minimum stock level.

For example:

100W Solar Panel

Minimum stock level: 5

If stock reaches:

4 units

the system will trigger a low-stock warning.

The administrator could see:

> **⚠ LOW STOCK**
> **100W Solar Panel**
> **Remaining: 4 units**
> **Minimum Level: 5 units**

This allows management to reorder before the product completely runs out.

3.9 Custom Stock Thresholds

Different products may require different minimum stock levels.

For example:

Product	Minimum Stock
Solar Cable	50
100W Panel	10
550W Panel	5
5kW Inverter	2
10kWh Battery	1

The Super Admin or authorised Inventory Manager can configure these thresholds according to the importance and sales frequency of each product.

3.10 Out-of-Stock Management

When available inventory reaches zero, the system will automatically mark the product as:

OUT OF STOCK

The product can remain visible on the website but customers will not be able to place a normal order for it.

Alternatively, the business can configure the product to:

- **Hide from storefront.**
- **Display "Out of Stock."**
- **Allow enquiry.**
- **Allow customers to request notification when available.**

3.11 Multiple Store and Warehouse Locations

The inventory system will support multiple physical locations.

For example:

Nairobi

Main warehouse and distribution centre.

Machakos

Retail shop and customer collection point.

Kitui

Regional retail location.

Each location will maintain its own stock balance.

Example:

Product	Nairobi	Machakos	Kitui	Total
550W Panel	30	15	10	55
5kW Inverter	8	3	2	13
10kWh Battery	4	2	1	7

This gives management visibility across the entire business.

3.12 Stock Transfers Between Locations

Authorised staff will be able to transfer products from one location to another.

Example:

Transfer Request

From: Nairobi Warehouse
To: Machakos Store
Product: 550W Solar Panel
Quantity: 10

The system will automatically record:

Nairobi:

30 → 20

Machakos:

15 → 25

The transfer history will include:

- **Sending location.**
- **Receiving location.**
- **Product.**
- **Quantity.**
- **Date.**
- **User initiating the transfer.**
- **User receiving the transfer.**
- **Transfer status.**
- **Reference number.**

3.13 Transfer Approval

For stronger internal control, the system can use an approval process.

For example:

Inventory Manager in Nairobi

creates:

> **Transfer Request #TR-0015**

The receiving location can confirm:

> **Received — 10 units**

This helps prevent discrepancies between the sending and receiving stores.

3.14 Stock Movement History

Every change to inventory will be recorded.

The history could show:

Date	Product	Movement	Quantity	Location	User

20/10/26	**550W Panel**	**Stock In**	**+20**	**Machakos**	**Inventory Manager**
20/10/26	**550W Panel**	**Sale**	**-3**	**Machakos**	**System**
20/10/26	**550W Panel**	**Transfer**	**-5**	**Machakos**	**Inventory Manager**
21/10/26	**550W Panel**	**Transfer In**	**+5**	**Nairobi**	**Inventory Manager**

This provides a complete audit trail.

3.15 Supplier Management

The inventory system will maintain supplier records.

Each supplier can have:

- **Supplier name.**
- **Company name.**
- **Contact person.**
- **Telephone.**
- **Email.**
- **Address.**
- **Products supplied.**
- **Buying prices.**
- **Payment terms.**
- **Supplier notes.**

Products can then be associated with their respective suppliers.

3.16 Supplier Product Pricing

The system can maintain the cost price of products.

For example:

550W Solar Panel

Supplier A: KES 28,000
Supplier B: KES 29,500
Supplier C: KES 27,800

This gives the business better visibility when evaluating suppliers.

Access to this information will be restricted because buying prices are commercially sensitive.

3.17 Purchase Records

The system can maintain records of stock received from suppliers.

For example:

Purchase Reference: PUR-2026-0091

Supplier: ABC Solar Ltd
Date: 20 October 2026
Product: 550W Panels
Quantity: 50
Unit Cost: KES 28,000
Total: KES 1,400,000

This creates a historical record of inventory acquisition.

3.18 Damaged Stock

Products that become damaged will be separated from sellable inventory.

For example:

Physical stock:

100 units

Damaged:

3 units

Available for sale:

97 units

The system will record:

- **Product.**
- **Quantity.**
- **Location.**
- **Date.**
- **Reason for damage.**
- **User reporting it.**
- **Notes.**
- **Supporting reference/photo where applicable.**

3.19 Lost Stock

Authorised users can record stock that is missing.

For example:

> **Product: Solar Floodlight**
> **Quantity: 2**
> **Location: Nairobi Warehouse**
> **Reason: Stock discrepancy**
> **Reported by: Inventory Manager**

The system can require appropriate authorisation before permanently adjusting stock.

3.20 Returned Products

Customers may return products according to the company's return policy.

The system will allow staff to record:

- **Original order.**
- **Customer.**
- **Product.**
- **Quantity.**
- **Reason for return.**
- **Condition.**
- **Return date.**
- **Action taken.**

The returned product can then be classified as:

- **Resalable.**
- **Damaged.**
- **Awaiting inspection.**
- **Warranty replacement.**
- **Supplier return.**

This prevents returned products from automatically being treated as normal available stock.

3.21 Stock Adjustment

Sometimes the physical stock count may differ from the system.

For example:

System stock:

50 units

Physical count:

48 units

An authorised user can create a stock adjustment.

The system will require:

- **Product.**
- **Previous quantity.**
- **New quantity.**
- **Difference.**
- **Reason.**
- **User.**
- **Date.**

This creates accountability rather than silently changing the quantity.

3.22 Physical Stocktaking

The system can support periodic physical stocktaking.

Staff can physically count products and compare them with the system.

For example:

System: 100
Physical: 98
Difference: -2

The discrepancy can then be investigated and, where approved, adjusted.

This is especially useful for businesses with large inventories.

3.23 SKU Management

Every product will have a unique Stock Keeping Unit (SKU).

Example:

SOL-PNL-550-MONO-001

SKU codes will help staff quickly identify products.

The system can use SKUs for:

- **Inventory tracking.**
- **Orders.**
- **Quotations.**
- **Invoices.**
- **Receipts.**
- **Stock transfers.**
- **Reports.**

3.24 Barcode Readiness

Products can be assigned barcode values.

A future or current barcode scanner can then be used to identify products quickly.

For example:

Scan → Product Identified → Price Displayed → Stock Updated

This can be particularly useful for a physical retail shop or warehouse.

3.25 Inventory Dashboard

The Inventory Manager will have a dedicated dashboard showing important information at a glance.

Example:

Inventory Overview

Total Products: 426

Products In Stock: 382

Low Stock: 27

Out of Stock: 17

Locations: 3

Stock Value: KES X,XXX,XXX

The dashboard can also show:

- **Recently received stock.**
- **Recent stock movements.**
- **Low-stock products.**
- **Out-of-stock products.**
- **Recent transfers.**
- **Damaged stock.**
- **Returned products.**

3.26 Inventory Reports

The system will provide reports that help management understand stock performance.

Stock Report

Shows current quantity of every product.

Low-Stock Report

Shows products below their minimum threshold.

Stock Movement Report

Shows every addition, sale, adjustment and transfer.

Stock Valuation Report

Shows the value of inventory based on authorised cost information.

Damaged Stock Report

Shows damaged products and associated quantities.

Lost Stock Report

Shows recorded stock losses.

Returned Stock Report

Shows customer returns and their status.

Location Report

Shows stock held at each physical location.

3.27 Inventory and E-Commerce Integration

The most important feature is that inventory will not operate separately from the online store.

The two systems will be connected.

For example:

Customer buys 3 panels online

↓

Order created

↓

Inventory checks availability

↓

3 units reserved/allocated according to order workflow

↓

Order confirmed

↓

Stock is deducted

↓

Product availability changes

↓

Customer sees updated availability

↓

Staff dashboard reflects the new quantity

This creates a continuous flow of information.

3.28 Inventory and Business Documents Integration

Inventory will also connect with Quotations, Invoices and Receipts.

For example:

Quotation

5 × 550W Solar Panels

↓

Customer accepts quotation

↓

Invoice generated

↓

Order confirmed

↓

Inventory updated

↓

Payment received

↓

Receipt generated

This reduces duplicate data entry.

3.29 Inventory Security

Not every employee will have permission to modify inventory.

For example:

Super Admin

Full inventory control.

Inventory Manager

Operational inventory control.

Sales Agent

Can see product availability but cannot directly change stock.

Customer

Can only see public availability such as:

- **In Stock**
- **Low Stock**
- **Out of Stock**

The API will enforce these permissions.

3.30 Inventory Audit Trail

Every sensitive inventory action will be traceable.

The system can answer:

- **Who added the stock?**
- **Who removed the stock?**
- **Who changed the quantity?**
- **Who transferred the stock?**
- **When did the change happen?**
- **Which location was involved?**
- **Why was the adjustment made?**

For example:

> **Stock Adjustment**
> **Product: 5kW Hybrid Inverter**
> **Previous: 8**
> **New: 7**
> **Difference: -1**
> **Reason: Damaged unit**
> **User: Inventory Manager**
> **Date: 20 October 2026**

This is important for accountability and internal controls.

3.31 Example End-to-End Inventory Scenario

Assume the company receives 100 units of 550W Solar Panels.

Step 1 — Stock Received

Nairobi Warehouse receives:

100 units

System records:

> **Stock = 100**

Step 2 — Transfer

20 units are transferred to Machakos.

System records:

Nairobi = 80
Machakos = 20

Step 3 — Online Sale

A customer buys 3 panels from Machakos.

System records:

Machakos = 17

Step 4 — Physical Sale

A customer purchases 5 panels directly from the Machakos shop.

System records:

Machakos = 12

Step 5 — Damage

One panel is damaged.

System records:

Available = 11
Damaged = 1

Step 6 — Low Stock

If the minimum stock level is 5, the system continues monitoring the remaining 11 units.

When stock falls to:

5 or below

the system generates a low-stock alert.

Step 7 — Replenishment

The business receives another 30 units.

System records:

Machakos: 11 + 30 = 41

The entire inventory history remains available.

4. Sales, Orders & Business Document Management

This section explains how the platform will manage the complete sales cycle, from the moment a customer places an order or a sales agent creates a quotation, up to invoicing, payment, receipt generation, delivery, and reporting.

4.1 Order Management

The system will provide a centralised order management module where all customer orders are recorded and managed.

Each order will contain:

- **Unique order number**
- **Customer name**
- **Customer phone number**
- **Customer email**
- **Delivery address**
- **Products ordered**
- **Product quantities**
- **Unit prices**
- **Discounts**
- **Subtotal**
- **VAT, where applicable**
- **Delivery charges**
- **Total amount**
- **Payment method**
- **Payment status**
- **Order status**
- **Date and time of order**
- **Sales agent responsible, where applicable**

This ensures that the business has one reliable record for every transaction.

4.2 Order Status Management

Orders will move through clearly defined stages.

A typical order workflow will be:

Pending → Confirmed → Processing → Packed → Dispatched → Delivered → Completed

Other statuses may include:

- **Cancelled**
- **Payment Failed**
- **Awaiting Payment**
- **Returned**
- **Refunded**

This allows staff to immediately understand the current state of an order.

For example:

> **Customer orders a 550W solar panel → order is received → stock is confirmed → payment is verified → product is packed → dispatched → delivered → order marked completed.**

4.3 Automatic Order Numbering

Every order will receive a unique reference number.

For example:

ORD-2026-000145

This number can be used by:

- **Customers**
- **Sales agents**
- **Inventory managers**
- **Administrators**

It makes it easier to search for transactions and resolve customer enquiries.

4.4 Sales Agent Order Processing

Sales agents will have a dedicated interface for handling sales.

A sales agent can:

1. **Search for a customer.**
2. **Select products.**
3. **Check available stock.**
4. **Add quantities.**
5. **Apply authorised discounts.**
6. **Generate a quotation.**
7. **Convert an approved quotation into an invoice.**
8. **Record payment.**
9. **Generate a receipt.**
10. **Update order status.**
11. **Mark the order as packed or delivered.**

The sales agent does not need to manually calculate totals or create documents using Microsoft Word or Excel.

The system performs these calculations automatically.

4.5 Quotation Management

The system will include a professional quotation generator.

A quotation can be created when a customer wants to know the cost of a solar installation or equipment before committing to purchase.

For example, a quotation could contain:

Item	Qty	Unit Price	Total
550W Solar Panel	4	KSh 35,000	KSh 140,000
5kVA Inverter	1	KSh 85,000	KSh 85,000
Solar Cables	1	KSh 8,000	KSh 8,000

Mounting Accessories	**1**	**KSh 12,000**	**KSh 12,000**

The system automatically calculates:

- **Subtotal**
- **Discount**
- **VAT**
- **Delivery charges**
- **Grand total**

The quotation will contain the company's:

- **Logo**
- **Business name**
- **Address**
- **Telephone number**
- **Email**
- **KRA PIN**
- **VAT information**
- **Quotation number**
- **Date**
- **Validity period**
- **Customer information**
- **Terms and conditions**

4.6 Quotation Numbering

Every quotation will automatically receive a unique number.

Example:

QUO-2026-000078

This eliminates duplicate quotation numbers and makes document tracking easier.

The system can also record:

- **Draft**
- **Sent**

- **Accepted**
- **Rejected**
- **Expired**
- **Converted to Invoice**

This provides a clear history of the quotation.

4.7 Converting a Quotation Into an Invoice

One of the important automation features is the ability to convert an approved quotation directly into an invoice.

For example:

Quotation

KSh 250,000

↓
Customer accepts

Convert to Invoice

↓

Invoice

KSh 250,000

The sales agent does not have to re-enter all products manually.

The system transfers:

- **Customer details**
- **Products**
- **Quantities**
- **Prices**
- **Discounts**
- **VAT**
- **Total**
- **Reference number**

This reduces errors and saves time.

4.8 Invoice Management

The platform will automatically generate professional invoices.

Each invoice will have a unique number such as:

INV-2026-000238

The invoice will include:

Business information

- **Company logo**
- **Business name**
- **Physical address**
- **Telephone**
- **Email**
- **KRA PIN**
- **VAT details**

Customer information

- **Customer name**
- **Phone**
- **Email**
- **Address**

Transaction information

- **Invoice number**
- **Invoice date**
- **Due date**
- **Order number**
- **Sales agent**

Product information

- **Product name**
- **SKU**
- **Quantity**
- **Unit price**

- **Discount**
- **VAT**
- **Total**

4.9 Invoice PDF Generation

Invoices will be generated automatically as professional PDF documents.

Users with appropriate permissions can:

- **Download invoice**
- **Print invoice**
- **Send via email**
- **Share through WhatsApp**
- **View invoice online**

For example:

INV-2026-000238.pdf

The PDF can be generated directly from the system without requiring staff to prepare documents manually.

4.10 Receipt Management

After payment has been confirmed, the system can generate an official receipt.

Example:

REC-2026-000193

The receipt will contain:

- **Receipt number**
- **Customer**
- **Invoice reference**
- **Payment amount**
- **Payment method**

- **Transaction/reference number**
- **Date**
- **Sales agent**
- **Company details**

Payment methods can include:

- **M-Pesa**
- **Bank**
- **Cash**
- **Other approved methods**

4.11 M-Pesa Payment Management

The system will support M-Pesa payment information as part of the checkout and sales process.

The basic implementation will display the business M-Pesa Till number to customers.

The customer can make payment and provide the relevant transaction reference.

The system can then associate the payment with the customer's order.

Optional STK Push

An automated M-Pesa Daraja STK Push integration can be added as an optional feature at the previously proposed additional cost of:

KES 10,000

With STK Push, the customer enters their phone number and receives an M-Pesa payment prompt directly on their phone.

4.12 Payment Status

Every order will have a payment status.

Examples:

- **Pending**
- **Awaiting Payment**
- **Paid**
- **Partially Paid**
- **Failed**
- **Refunded**

This is particularly useful for larger solar equipment purchases where customers may make deposits or partial payments.

For example:

Order Total: KSh 300,000
Amount Paid: KSh 150,000
Balance: KSh 150,000

The system can maintain this information against the order.

4.13 Partial Payment Management

For approved business processes, the system can support partial payments.

Example:

A customer purchases a solar system worth:

KSh 500,000

They pay:

KSh 200,000 deposit

The system records:

- **Invoice total: KSh 500,000**
- **Paid: KSh 200,000**
- **Outstanding: KSh 300,000**

Additional payments can then be recorded until the invoice is fully settled.

4.14 Automatic Financial Calculations

The system will automatically calculate:

Subtotal

`Quantity × Unit Price`

Then:

Subtotal - Discount + Applicable VAT + Delivery = Grand Total

This eliminates manual calculations by sales staff.

It also reduces common errors when handling multiple products.

4.15 Discounts

Authorised users can apply discounts to products or entire orders.

For example:

Product price:

KSh 50,000

Discount:

KSh 5,000

Customer pays:

KSh 45,000

The system records the discount so management can later analyse how much the business has given away through discounts.

Discount permissions can be restricted according to user role.

4.16 Customer Purchase History

Each registered customer will have a transaction history.

The business can view:

- **Previous orders**
- **Products purchased**
- **Amount spent**
- **Invoices**
- **Receipts**
- **Payment history**
- **Outstanding balances**
- **Delivery history**

This is useful for repeat customers and corporate clients.

4.17 Sales Dashboard

The administration dashboard will provide a summary of business performance.

Key indicators can include:

Today's Sales

Total sales generated today.

Monthly Sales

Total sales generated during the current month.

Orders Today

Number of orders received today.

Pending Orders

Orders awaiting processing.

Completed Orders

Successfully delivered/completed orders.

Outstanding Payments

Invoices or orders with unpaid balances.

Top-Selling Products

Products generating the highest sales volume.

Low-Stock Products

Products approaching their stock threshold.

4.18 Sales Reports

The system will generate reports that help management understand business performance.

Reports may include:

- **Daily sales report**
- **Weekly sales report**
- **Monthly sales report**
- **Annual sales report**
- **Product sales report**
- **Sales-agent performance**
- **Customer purchase report**
- **Payment report**
- **Outstanding invoice report**
- **Discount report**
- **Cancelled orders**
- **Returned orders**
- **Delivery report**

Reports can be filtered by:

- **Date**
- **Product**
- **Category**
- **Sales agent**

- **Customer**
- **Location**
- **Payment method**

4.19 Sales Agent Performance

Management can monitor sales performance by individual sales agent.

For example:

Sales Agent	Orders	Sales
Agent A	**45**	**KSh 2.4M**
Agent B	**31**	**KSh 1.8M**
Agent C	**22**	**KSh 1.2M**

This helps management identify:

- **High-performing staff**
- **Sales trends**
- **Areas requiring support**
- **Sales targets**

Sensitive financial information can remain restricted to authorised administrators.

4.20 Business Document Management

The platform will bring the major sales documents into one system:

Quotation → Invoice → Receipt

This creates a complete transaction trail.

For example:

QUO-2026-000078

↓

Customer accepts

↓

INV-2026-000238

↓

Customer pays

↓

REC-2026-000193

This makes the system much more professional than managing documents separately in Word, Excel, or physical files.

4.21 Automatic Document Numbering

The system will automatically generate unique document numbers for:

- **Quotations**
- **Invoices**
- **Receipts**
- **Orders**
- **Stock movements**
- **Transfers**

This prevents duplicate numbering and simplifies document tracking.

4.22 WhatsApp and Email Sharing

Generated quotations, invoices and receipts can be downloaded and shared with customers.

The platform can support workflows where staff send documents through:

- **WhatsApp**
- **Email**

For example, after generating an invoice, a sales agent can download the PDF and send it directly to the customer.

4.23 Document Search

Staff will be able to search for documents using references such as:

INV-2026-000238

or customer information such as:

- **Customer name**
- **Phone number**
- **Email**
- **Order number**

This makes retrieval much faster than searching through physical files.

4.24 Order Cancellation

Authorised staff can cancel an order where necessary.

The system should record:

- **Order number**
- **Cancellation reason**
- **User who cancelled it**

- **Date/time**
- **Payment status**
- **Stock impact**

If stock had already been reserved, the system can release the reserved quantity back into available inventory according to the business rules.

4.25 Returns and Refunds

Where a product is returned, the system can connect the return to the original order.

It can record:

- **Customer**
- **Original order**
- **Product**
- **Quantity returned**
- **Reason**
- **Condition**
- **Date**
- **Staff member**
- **Refund status**

The returned product can then be classified as:

- **Resalable**
- **Damaged**
- **Awaiting inspection**
- **Warranty case**
- **Supplier return**

This connects directly with the inventory system explained in Part 3.

4.26 Integration With Inventory

Sales and inventory will not operate as separate systems.

For example:

Available Stock = 20

Customer purchases:

3 units

System automatically updates:

Available Stock = 17

If an order is cancelled before dispatch, the system can release the reserved stock.

This ensures that customers and staff see accurate stock availability.

4.27 Integration With Customer Accounts

Customers will be able to access their own transaction history.

Inside their account they can view:

- **Orders**
- **Order status**
- **Quotations, where applicable**
- **Invoices**
- **Receipts**
- **Payment information**
- **Delivery information**

A customer cannot access another customer's records.

4.28 Security and Permissions

Sales and document operations will be controlled by user roles.

Super Admin

Full access.

Inventory Manager

Operational access to orders, inventory and business documents, subject to restricted financial/system settings.

Sales Agent

Sales-related access only.

Customer

Own orders and documents only.

The system will enforce these permissions through the backend API, not merely by hiding buttons on the frontend.

4.29 Audit Trail

Important actions will be recorded.

For example:

> **Agent John converted QUO-2026-000078 into INV-2026-000238 on 20 October 2026 at 14:35.**

Another example:

> **Inventory Manager Mary changed order status from Processing to Packed.**

This creates accountability and makes it easier to investigate mistakes or disputes.

4.30 Complete Sales Workflow

The complete process will work as follows:

Customer visits website

↓

Searches for solar product

↓

Views product details

↓

Adds product to cart

↓

Checks out

↓

Selects payment method

↓

Order created

↓

Stock reserved

↓

Payment confirmed

↓

Invoice generated

↓

Warehouse prepares order

↓

Stock deducted

↓

Order dispatched

↓

Customer receives product

↓

Receipt generated

↓

Order marked Completed

↓

Transaction appears in sales reports

This creates a complete digital trail from customer enquiry to completed sale.

5. Customer Management & Customer Experience System

The **Customer Management & Customer Experience System** will manage the complete relationship between the business and its customers. It will allow customers to create accounts, browse products, place orders, track purchases, access documents, communicate with the business and maintain their personal information.

At the same time, authorised staff will have a central customer-management interface where they can view customer profiles, purchase history, orders, payments and interactions.

5.1 Customer Registration

Customers will be able to create an account on the platform.

The registration process can capture:

- Full name
- Phone number
- Email address
- Password
- Delivery address
- County
- Town/area
- Optional company/business name

- Optional KRA PIN for business customers

Once registered, the customer will have access to their personal dashboard.

Example

A customer registering as:

Name: John Mwangi
Phone: 07XXXXXXXX
Email: john@example.com
Location: Machakos

will have a personal account where all his future orders and documents are maintained.

5.2 Customer Login

Registered customers will securely log into their accounts.

The login system will allow customers to access:

- Previous orders
- Current orders
- Invoices
- Receipts
- Wishlist
- Saved addresses
- Account information
- Product reviews
- Order tracking

Customers will only be able to access information belonging to their own account.

5.3 Customer Dashboard

Each customer will have a personalised dashboard.

The dashboard can display:

My Orders

Number of orders placed.

Pending Orders

Orders still being processed.

Completed Orders

Successfully delivered purchases.

Outstanding Payments

Unpaid or partially paid transactions.

Wishlist

Products saved for future purchase.

Recent Purchases

Products recently bought.

Documents

Available invoices and receipts.

A simplified dashboard could look like:

Welcome, John

Information	Value
Total Orders	8
Pending Orders	2
Completed Orders	6
Wishlist Items	5
Outstanding Balance	KSh 25,000

5.4 Customer Profile Management

Customers will be able to update their personal information.

They can manage:

- Name
- Phone number
- Email
- Password
- Delivery address
- Town
- County
- Company information

Changes to sensitive information can require additional verification where necessary.

5.5 Multiple Delivery Addresses

The system can allow customers to save more than one delivery address.

For example:

Address 1 — Home
Machakos Town

Address 2 — Office
Nairobi CBD

During checkout, the customer can select the appropriate address.

This is especially useful for customers purchasing solar equipment for different sites.

5.6 Customer Types

The system can support different types of customers.

For example:

Individual Customer

A person buying solar equipment for their home.

Business Customer

A company purchasing solar equipment for business operations.

Contractor/Installer

A solar technician purchasing equipment for multiple projects.

Institution

Schools, churches, hospitals, NGOs and other organisations.

This allows the business to understand its customer base.

5.7 Customer Purchase History

Every customer's transactions will be stored under their account.

The system can display:

- Order number
- Date
- Products purchased
- Quantity
- Amount
- Payment status
- Order status
- Delivery status

Example:

Order	Date	Amount	Status
ORD-00125	10/10/2026	KSh 85,000	Delivered
ORD-00147	15/10/2026	KSh 120,000	Processing

ORD-00158	19/10/2026	KSh 45,000	Pending Payment

This creates a complete customer transaction history.

5.8 Customer Order Tracking

Customers will be able to track their orders without calling the business for every update.

For example:

Order Received ✓
↓
Payment Confirmed ✓
↓
Processing ✓
↓
Packed ✓
↓
Dispatched
↓
Delivered

The current status will be clearly displayed.

5.9 Customer Notifications

The system can provide notifications relating to important customer activities.

Notifications may include:

- Order received
- Payment confirmed
- Order processing
- Order packed
- Order dispatched
- Order delivered
- Invoice generated

- Payment reminder
- Order cancellation
- Return/refund update

Notifications can be displayed inside the customer's account and, where configured, sent through email, SMS or WhatsApp integrations.

5.10 Invoice Access

Customers will be able to access invoices relating to their purchases.

For example:

INV-2026-000238

The customer can:

- View invoice
- Download PDF
- Print invoice
- Keep the invoice for accounting purposes

This eliminates the need for customers to repeatedly request copies from staff.

5.11 Receipt Access

Once payment has been recorded, the customer can access their receipt.

Example:

REC-2026-000193

The receipt can contain:

- Receipt number
- Invoice number
- Customer name
- Amount paid
- Payment method

- Transaction reference
- Date
- Business details

5.12 Product Datasheet Access

Solar products often require technical specifications.

Customers will therefore be able to access product datasheets where provided.

A product page may contain:

550W Solar Panel

- Power: 550W
- Voltage
- Current
- Dimensions
- Weight
- Efficiency
- Warranty
- Manufacturer
- Datasheet PDF

The customer can download the technical datasheet before making a purchase.

5.13 Wishlist

Customers will be able to save products they are interested in.

For example:

❤️ 5KW Hybrid Inverter
❤️ 10KWh Lithium Battery
❤️ 550W Solar Panel

The customer can return to the wishlist later and purchase the product.

This is particularly useful when customers are comparing products or waiting until they have sufficient funds.

5.14 Product Comparison

Customers will be able to compare products before purchasing.

For example, they could compare four solar panels based on:

- Price
- Wattage
- Brand
- Efficiency
- Warranty
- Dimensions
- Availability

This helps customers make informed purchasing decisions.

5.15 Product Reviews and Ratings

After purchasing a product, customers can be allowed to leave a review.

A review can contain:

- Star rating
- Written feedback
- Optional image

For example:

★★★★★

> "The solar panel was delivered on time and works as expected."

Reviews can help future customers understand the experiences of previous buyers.

5.16 Review Moderation

Reviews will not automatically appear publicly without appropriate controls.

Authorised staff can:

- View reviews
- Approve reviews
- Hide inappropriate reviews
- Remove spam
- Respond to customer feedback

This protects the quality and professionalism of the platform.

5.17 Customer Search for Staff

The back office will provide a customer management module.

Staff with appropriate permissions can search customers using:

- Name
- Phone number
- Email
- Customer ID
- Order number
- Company name

This makes customer support much faster.

5.18 Customer Profile for Staff

An authorised staff member can open a customer profile and see relevant business information.

For example:

Customer

John Mwangi

Contact

07XXXXXXXX

Orders

12

Total Purchases

KSh 850,000

Pending Orders

2

Outstanding Balance

KSh 50,000

Last Purchase

19 October 2026

This gives staff a complete picture of the customer relationship.

5.19 Customer Segmentation

The system can categorise customers according to business requirements.

Possible categories include:

- New Customer
- Returning Customer
- Regular Customer
- Contractor
- Business
- Institution
- Wholesale Customer

This can later support targeted marketing and customer retention.

5.20 Wholesale and Repeat Customers

The platform can be structured to support customers who purchase large quantities.

For example, a solar installer may regularly purchase:

- 50 panels
- 10 inverters
- 20 batteries
- Cables and accessories

The business can maintain the customer's transaction history and provide appropriate quotations or negotiated pricing through authorised staff.

5.21 Customer Communication

The system can provide several communication channels.

WhatsApp

A WhatsApp button can allow customers to quickly contact the business.

Email

Important documents and notifications can be shared through email.

Phone

The platform can display the business telephone number.

This creates multiple channels for customer support.

5.22 WhatsApp Sales Support

WhatsApp can also become part of the sales journey.

For example:

Customer:
"I need a 5KVA inverter."

The customer can click the WhatsApp button and communicate directly with the sales team.

The business can then:

1. Understand the customer's requirements.
2. Recommend suitable products.
3. Prepare a quotation.
4. Send the quotation.
5. Process the order.

This is especially useful because many Kenyan customers already use WhatsApp as a primary communication channel.

5.23 Customer Enquiry Management

Customers may have questions before purchasing.

Examples:

- Is the product available?
- What is the warranty?
- Can you deliver to Kitui?
- Which inverter works with this battery?
- Do you install solar systems?
- What is the delivery cost?

The platform can provide clear contact options and product information to reduce unnecessary enquiries.

5.24 Customer Account Security

Customer accounts will be protected using standard security controls.

These include:

- Password protection
- Secure authentication
- Encrypted HTTPS communication
- Server-side permission checks
- Session management
- Password reset functionality
- Protection against unauthorised access

Customer passwords will not be exposed to staff.

5.25 Password Reset

If a customer forgets their password, they can initiate a password recovery process.

The system can use the registered email or phone number to verify the customer and allow them to create a new password.

5.26 Customer Data Privacy

Customer information will be accessible according to the user's role and permissions.

A customer should only be able to see:

Their own information

while staff can only access information necessary for their assigned responsibilities.

For example:

Sales Agent: Customer contact and sales information.

Inventory Manager: Customer/order information needed for fulfilment.

Super Admin: Full authorised customer management.

5.27 Customer Support History

Where implemented, the system can maintain a history of customer interactions.

This can help staff understand:

- Previous enquiries
- Previous orders
- Complaints
- Returns
- Warranty cases
- Follow-ups

This reduces the problem of customers having to explain their entire history every time they contact the business.

5.28 Customer Returns and Warranty

Customers can initiate enquiries regarding returned products or warranty claims.

The system can associate the claim with:

- Customer
- Order
- Invoice
- Product
- Purchase date
- Serial number, where applicable
- Warranty status

This is particularly important for solar equipment such as:

- Inverters
- Batteries
- Solar panels
- Water pumps
- Charge controllers

5.29 Customer Loyalty and Future Expansion

The architecture can allow future implementation of customer loyalty features such as:

- Loyalty points
- Discount codes
- Customer-specific offers
- Referral programmes
- Promotional campaigns
- Birthday/anniversary offers
- VIP customer categories

These can be introduced later without redesigning the entire platform.

5.30 Customer Experience on Mobile

The platform will be designed to work properly on smartphones.

Customers should be able to:

- Browse products
- Search
- Compare
- Add to cart
- Checkout
- Make payment
- Track orders
- Download invoices
- Contact the business

from a mobile phone.

This is particularly important because a significant portion of e-commerce traffic is expected to come from mobile devices.

5.31 Customer Management Dashboard

The administration dashboard can provide customer statistics.

For example:

Metric	Value
Total Customers	1,250
New This Month	185
Returning Customers	720
Active Customers	840
Business Customers	95
Pending Orders	37

This gives management visibility into customer growth.

5.32 Customer Analytics

The system can help management understand purchasing behaviour.

Reports can identify:

- Most active customers
- Highest-value customers
- Most purchased products
- Repeat purchase rate
- Average order value
- Customer growth
- Customer location
- Product preferences

For example, management may discover that customers from Machakos frequently purchase solar batteries while customers from Nairobi purchase more inverters.

This information can support purchasing and marketing decisions.

5.33 Customer-to-Sales Integration

Customer management will be directly connected to the sales module.

The workflow becomes:

Customer

↓

Product browsing

↓

Cart

↓

Order

↓

Payment

↓

Invoice

↓

Delivery

↓

Receipt

↓

Customer history

Every transaction becomes part of the customer's account.

5.34 Customer-to-Inventory Integration

The customer module will also connect to inventory.

When a customer views a product, the platform can show:

In Stock

or

Only 3 Left

or

Out of Stock

This prevents customers from placing orders for products that are unavailable.

5.35 Customer Experience Example

Consider a customer named **Peter** who wants to purchase a solar system.

Step 1 — Registration

Peter creates an account.

Step 2 — Search

He searches for:

5KVA Solar Inverter

Step 3 — Comparison

He compares three available inverters.

Step 4 — Product Selection

He selects one and adds it to his cart.

Step 5 — Checkout

He enters his delivery address.

Step 6 — Payment

He selects M-Pesa.

Step 7 — Order

The system creates:

ORD-2026-00125

Step 8 — Invoice

The system generates:

INV-2026-00201

Step 9 — Fulfilment

The inventory team prepares the product.

Step 10 — Delivery

The order status changes to:

Delivered

Step 11 — Receipt

The system generates:

REC-2026-00186

Step 12 — Review

Peter can rate the product and leave feedback.

His entire transaction remains available in his customer account.

5.36 Business Benefits

The Customer Management System provides the business with several advantages.

Better customer service

Staff can quickly access customer information and order history.

Increased customer trust

Customers can access their own invoices, receipts and order information.

More repeat purchases

Wishlists, purchase history and personalised experiences encourage customers to return.

Faster support

Staff do not need to search through physical files to find customer transactions.

Better sales decisions

Customer analytics show what customers are buying.

Professional image

A structured customer portal makes the company appear more established and organised.

Reduced administrative work

Documents, orders and customer information are automatically linked.

Improved accountability

Every important customer-related action can be recorded.

5.37 Overall Customer Management Architecture

The customer experience will connect directly with the rest of the platform:

Customer

↓

Registration / Login

↓

Product Catalogue

↓

Search & Comparison

↓

Wishlist / Cart

↓

Checkout

↓

Payment

↓

Order Management

↓

Inventory

↓

Invoice

↓

Delivery

↓

Receipt

↓

Customer History

↓

Review / Repeat Purchase

This makes the customer module an important part of the entire **Jumia-style solar e-commerce ecosystem**, ensuring that the platform does not only sell products but also maintains a structured, professional and long-term relationship with every customer.

6. User Roles & Permissions

The system will use **role-based access control (RBAC)**. Each user will only access the functions required for their responsibilities. Permissions will be enforced at the backend/API level, ensuring that users cannot bypass restrictions simply by changing the frontend.

6.1 Super Admin / Business Owner

Access Level: 100% — Full System Control

The Super Admin is the highest-level user and has complete authority over the platform, business configuration and sensitive financial information.

Responsibilities and permissions:

- Access the complete administrative dashboard.
- Create, edit, deactivate and remove system users.
- Assign roles and permissions to users.
- Add, edit and delete products.
- Change product selling prices and discounts.
- Manage product categories and brands.
- Manage product images and technical datasheets.
- Create and manage product SKUs and barcodes.
- View and manage stock across all locations.
- Add, adjust, transfer and remove stock.
- View complete stock movement history.
- Manage suppliers and supplier purchasing information.
- View buying prices and selling prices.
- View gross and net profit information.
- View complete sales and financial reports.
- Manage customers.
- View and manage all orders.
- Approve, reject, cancel or modify orders where appropriate.
- Generate Quotations, Invoices and Receipts.
- Archive or permanently delete business documents where permitted.
- Configure company information.
- Configure company logo.
- Configure KRA PIN and VAT details.

- Configure M-Pesa Till/payment information.
- Configure low-stock thresholds.
- Manage store/warehouse locations.
- View system activity logs.
- Manage backups and data recovery.
- Access all system settings.
- Monitor the overall performance of the online store.

Restrictions:

- None within the agreed system scope.

The Super Admin effectively acts as the **business owner/system administrator** and retains final control over sensitive business information.

6.2 Inventory Manager / Operations Manager

Access Level: Approximately 95% — Operational Administration

The Inventory Manager is the primary day-to-day administrator responsible for keeping products, stock, suppliers and orders running smoothly.

This role is particularly useful for a staff member operating from a **Machakos, Kitui or Nairobi shop/store**.

Responsibilities and permissions:

Product Management

- Add new solar products.
- Edit product information.
- Delete products where permitted.
- Upload product images.
- Upload product datasheets.
- Assign products to categories.
- Assign brands.
- Set product specifications.
- Manage SKUs.
- Manage barcode information.
- Update product availability.

Inventory Management

- Add incoming stock.

- Adjust stock quantities.
- Record damaged stock.
- Record lost stock.
- Record returned stock.
- Transfer stock between locations.
- View current stock levels.
- View stock movement history.
- Receive low-stock alerts.
- Identify products that need restocking.

Supplier Management

- Add suppliers.
- Edit supplier information.
- View supplier contacts.
- Record supplier purchasing information.
- Associate suppliers with products.
- Track buying prices.

Order Management

- View customer orders.
- Approve orders.
- Reject orders.
- Update order status.
- Process orders for fulfilment.
- Mark orders as packed.
- Mark orders as dispatched/delivered.
- Handle returned orders.

Business Documents

- Create Quotations.
- Convert Quotations into Invoices.
- Generate Receipts.
- Download PDFs.
- Share documents through WhatsApp or email.
- View historical business documents.
- Archive documents.

Customer Management

- View customer accounts.
- View customer order history.
- Update customer information where permitted.
- Assist with customer orders and enquiries.

Reporting

- View sales reports.
- View order reports.
- View inventory reports.
- View low-stock reports.
- Monitor daily sales activity.

Restrictions:

The Inventory Manager cannot:

- Add or remove Super Admins.
- Delete system users.
- Change the company KRA PIN.
- Change the official company logo.
- Change M-Pesa Till/payment configuration.
- View sensitive net-profit/margin information based on buying versus selling price.
- Permanently delete historical invoices.
- Change critical system-level settings.
- Perform permanent deletion of protected business records.

Purpose of the role:

The Inventory Manager can essentially **run the store without constantly involving the owner**, while sensitive ownership and financial controls remain with the Super Admin.

6.3 Sales Agent / Sales Officer

Access Level: Approximately 40% — Sales Operations

The Sales Agent is focused primarily on **selling products and serving customers**, rather than managing the underlying business or inventory.

Responsibilities and permissions:

Customer Sales

- View available products.
- Search for products.
- Check product selling prices.
- Assist customers with product selection.
- View customer information required for sales.
- View customer orders.

Quotations

- Create Quotations for customers.
- Add products to Quotations.
- Specify quantities.
- Apply approved pricing.
- Download Quotations.
- Share Quotations through WhatsApp/email.

Invoices

- Convert an approved Quotation into an Invoice.
- Generate customer invoices.
- Download invoices.
- Share invoices with customers.

Receipts

- Generate Receipts for completed payments.
- Download receipts.
- Share receipts with customers.

Order Processing

- View incoming orders.
- Update permitted order statuses.
- Mark orders as **Packed**.
- Mark orders as **Dispatched**.
- Mark orders as **Delivered**, where authorised.

Customer Support

- Assist customers with order enquiries.
- Check order status.
- Help customers obtain their documents.

Restrictions:

The Sales Agent cannot:

- Add new products.
- Delete products.
- Edit core product information.
- Change product selling prices unless specifically authorised.
- View supplier buying prices.
- View profit margins.

- Adjust stock quantities.
- Transfer stock between locations.
- Manage suppliers.
- Change company settings.
- Change KRA PIN.
- Change M-Pesa settings.
- Manage users.
- Access sensitive financial reports.
- Permanently delete invoices or other protected records.

Purpose of the role:

The Sales Agent has enough access to **complete a sale from quotation through invoicing and receipt generation**, while preventing access to sensitive stock costs, profits and system configuration.

6.4 Customer

Access Level: Storefront Only

The Customer is an external user who interacts with the public e-commerce platform.

Customers have no access to the administrative system.

Customer capabilities:

Product Discovery

- Browse the homepage.
- Browse product categories.
- Search for products.
- Filter products by:
    - Price
    - Brand
    - Wattage
    - Category
    - Availability
- View product specifications.
- View product images.
- Download available datasheets.
- View related products.
- View product ratings and reviews.

Shopping

- Add products to cart.
- Modify quantities.
- Remove products from cart.
- Add products to wishlist.
- Compare up to four products.
- View product availability.

Checkout

- Place orders.
- Select payment method.
- Use the available M-Pesa payment option.
- Select bank payment where applicable.
- Select Cash on Delivery where applicable.
- Provide delivery information.
- Confirm orders.

Customer Account

- Register an account.
- Log in securely.
- Update permitted profile information.
- View previous orders.
- Track current orders.
- View order status.
- Download their own invoices.
- Download their own receipts.
- Download product datasheets.

Product Reviews

- Rate purchased products.
- Submit product reviews subject to system rules.

Customer Support

- Contact the business through the integrated WhatsApp button.
- Make enquiries about products and orders.

Restrictions:

Customers cannot:

- Access the admin dashboard.
- Access inventory.
- Change stock.

- View buying prices.
- View profit margins.
- View other customers.
- View supplier information.
- Create business invoices for other customers.
- Change company settings.
- Manage products.
- Manage orders belonging to other customers.
- Access financial reports.
- Manage users.